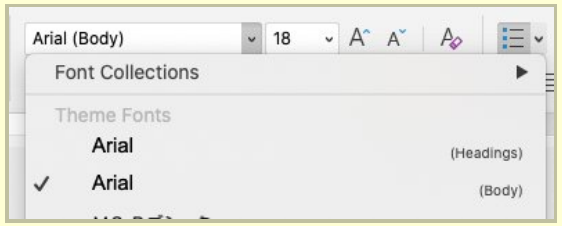
>>>>> Before you begin <<<<<

Capitalization guidelines

- Use **title case** for presentation titles
  - All words are capitalized except for prepositions, conjunctions, and articles
- Use **sentence case** for slide titles, section titles, and bullets
  - Just the first word in a text string is capitalized
- Always **capitalize proper nouns**, including Esri product names as noted in the Product Naming Guide

Font guidelines

- Always pick **Arial** [Headings] or [Body] from the **top** of the font list under Theme Fonts



>>>>> Before you begin <<<<<



Less is more

- Avoid paragraphs and long sentences
- Use the 5x5 rule — max of 5 lines, 5 words each
- One idea per slide



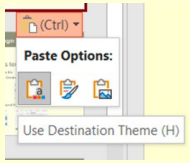
Presentation structure

- Content in longer presentations should be grouped using the following structure:
  - Sections > Topics > Subtopics
  - Note: A section may include multiple topics

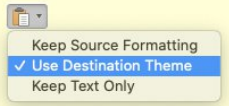
>>>>> Before you begin <<<<<

How to apply this template

1. Copy existing slides into this file
2. When pasting, choose Use Destination Theme



Windows

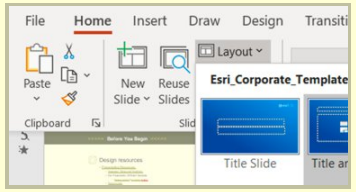


Mac

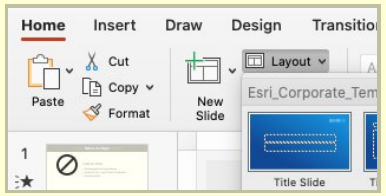


Web

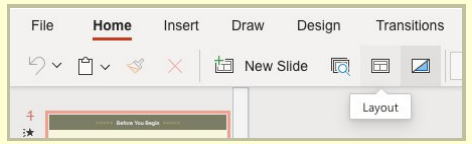
3. For consistent title placement, you may need to select a layout from the Layout menu



Windows



Mac



Web

>>>>> Before you begin <<<<<

Design resources

- Icons
  - From the Home ribbon above, click Esri Icons

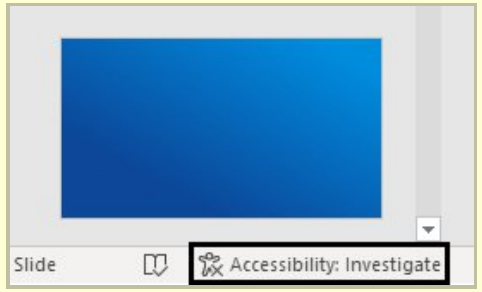


- Presentation Resources [link]
  - Diagrams, Tables and Timelines [link]
  - Esri Design Template [link]
  - Globes [link]
  - Backgrounds [link]
- Strategic Presentations [link]
- Stock photos
  - Adobe Stock images are available. See the details
  - **Do not use** stock images from Microsoft.

>>>>> Before you begin <<<<<

Accessibility [link]

- Check accessibility while designing slides with the Review tab or in the lower left-hand corner
- Add **alternative text** and mark graphical images as decorative
- Review slide **reading order**
- Create **unique slide titles**





Maps that do the talking

Data-driven visualization with MapLibre

George Owen

Agenda

Goal: Show **data-driven visualization** in the browser

- Open data sources
- MapLibre style specification
- Maps!
- Cartographic techniques

Open data sources

Spatial data fundamentals

Basic context

- Features
  - Geometry + Attributes
- Today: Using Vector Tiles
- Many other formats available

- Data-driven styles



Data sources

Focus of today's talk

- Today's study area: **City of Sacramento**

- Data sources used in today's talk:
  - City of Sacramento Open Data Hub
  - Sacramento County Open Data Hub
  - ArcGIS Living Atlas, Esri Demographics*
- Data methodology:
  - Started as feature services
  - Prepared, clipped, and enriched
  - Final rev published as vector tiles

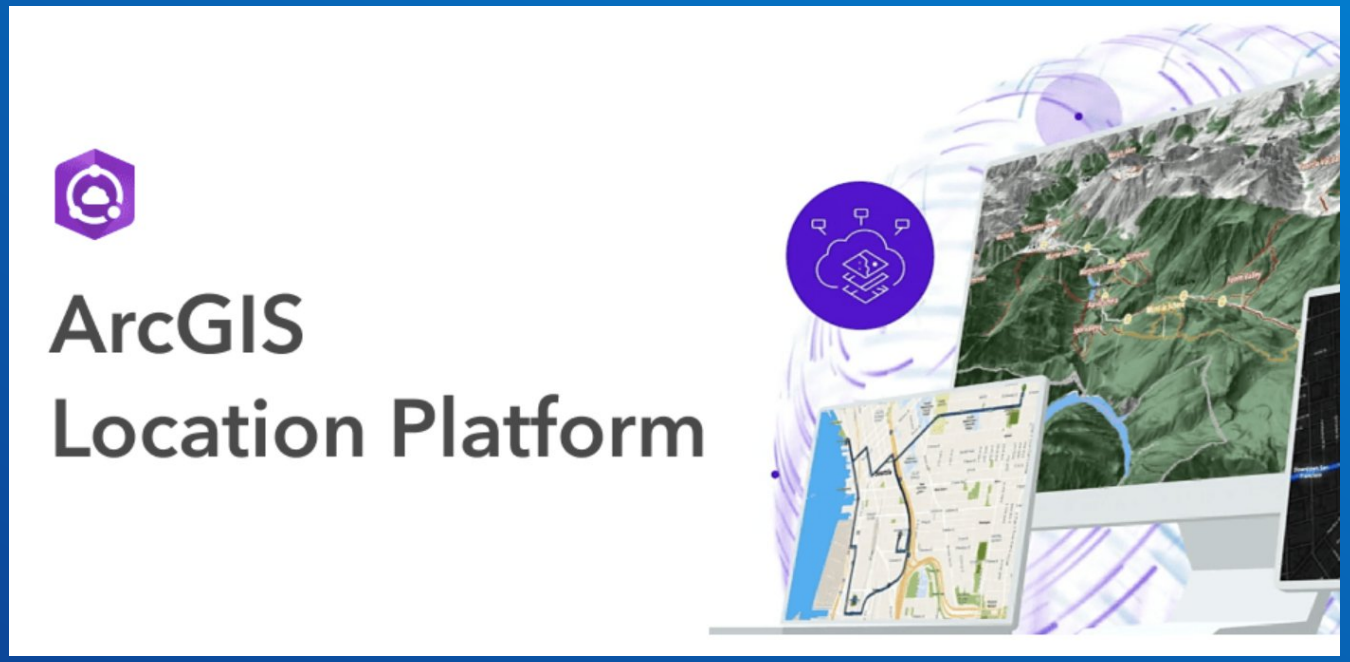


Accessing open data with ArcGIS

Use an ArcGIS account

- Free and open source
- Make a free account: ArcGIS Location Platform
  - Online and Enterprise also work
- Also gets data hosting, platform services

- location.arcgis.com



Style specification

MapLibre Style Specification

An open standard for visualization

- Open-source fork of Mapbox v1 style spec
- Part of MapLibre project
- Published in 2020 (ancient by today's standards)

- Widely supported, de facto interop standard
  - *Also note the OGC styles standard



Style document layout

An open standard for visualization

- A single JSON document containing:
  - Data sources
  - Style layers
  - Expressions
  - Fonts, glyphs, sprites

- MapLibre Style Specification

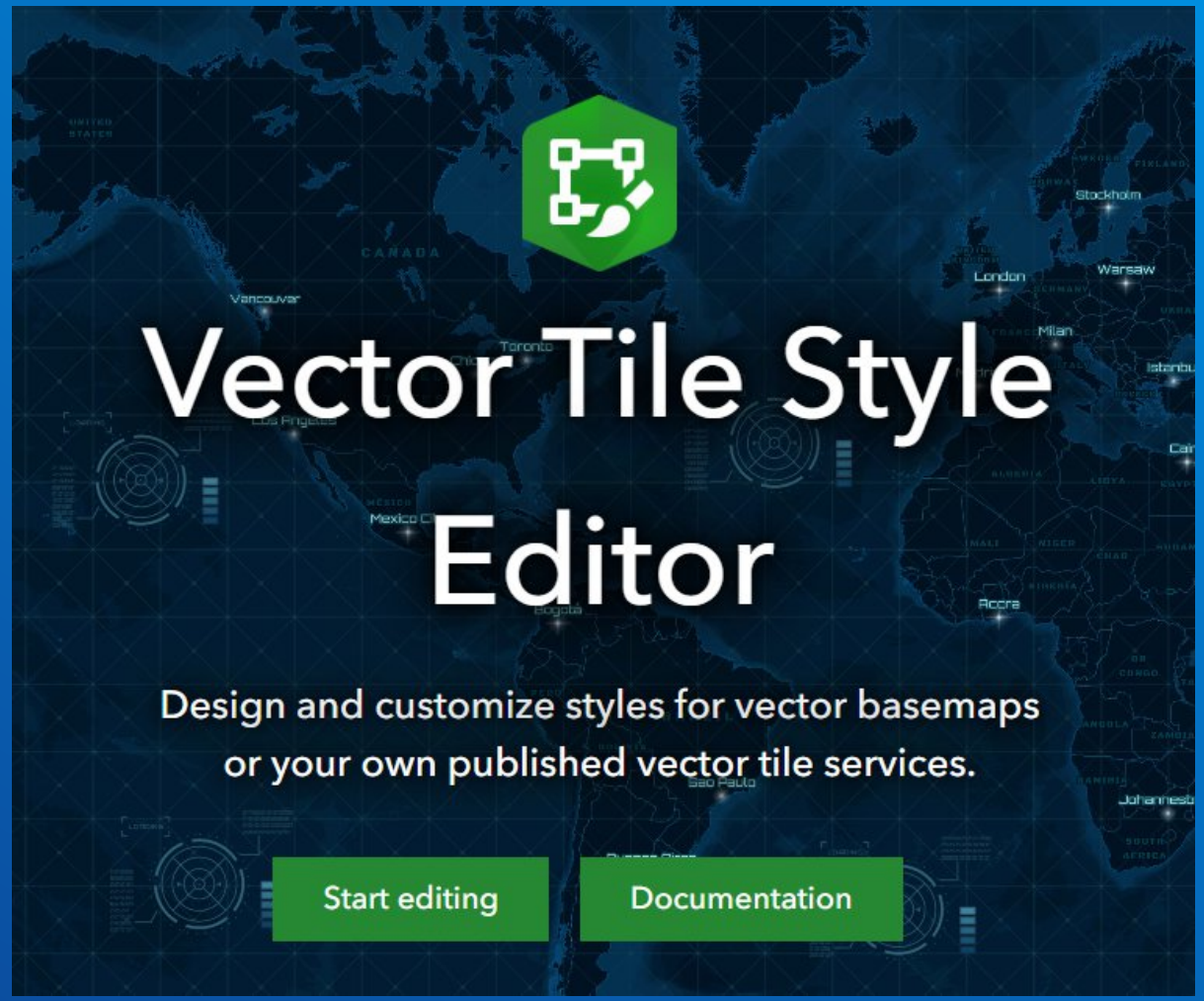
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        "coalesce",
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        ],
        ""
      ],
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        "A2",
        "A3",
        "A4",
        "A5",
        "A6"
      ],
      "#219AD2",
      [
        "B1",
        "B2",
        "B3"
      ],
```

Creating style JSON

MapLibre Style Spec can do anything!

- Recommend using web tools for sanity

- Vector Tile Style Editor
  - GUI-based approach to styling
  - Many layers at once
- Agentic workflows
  - Select colors, symbols, etc
  - Agentically generate style document using using a custom agent skill



Let’s see some maps!

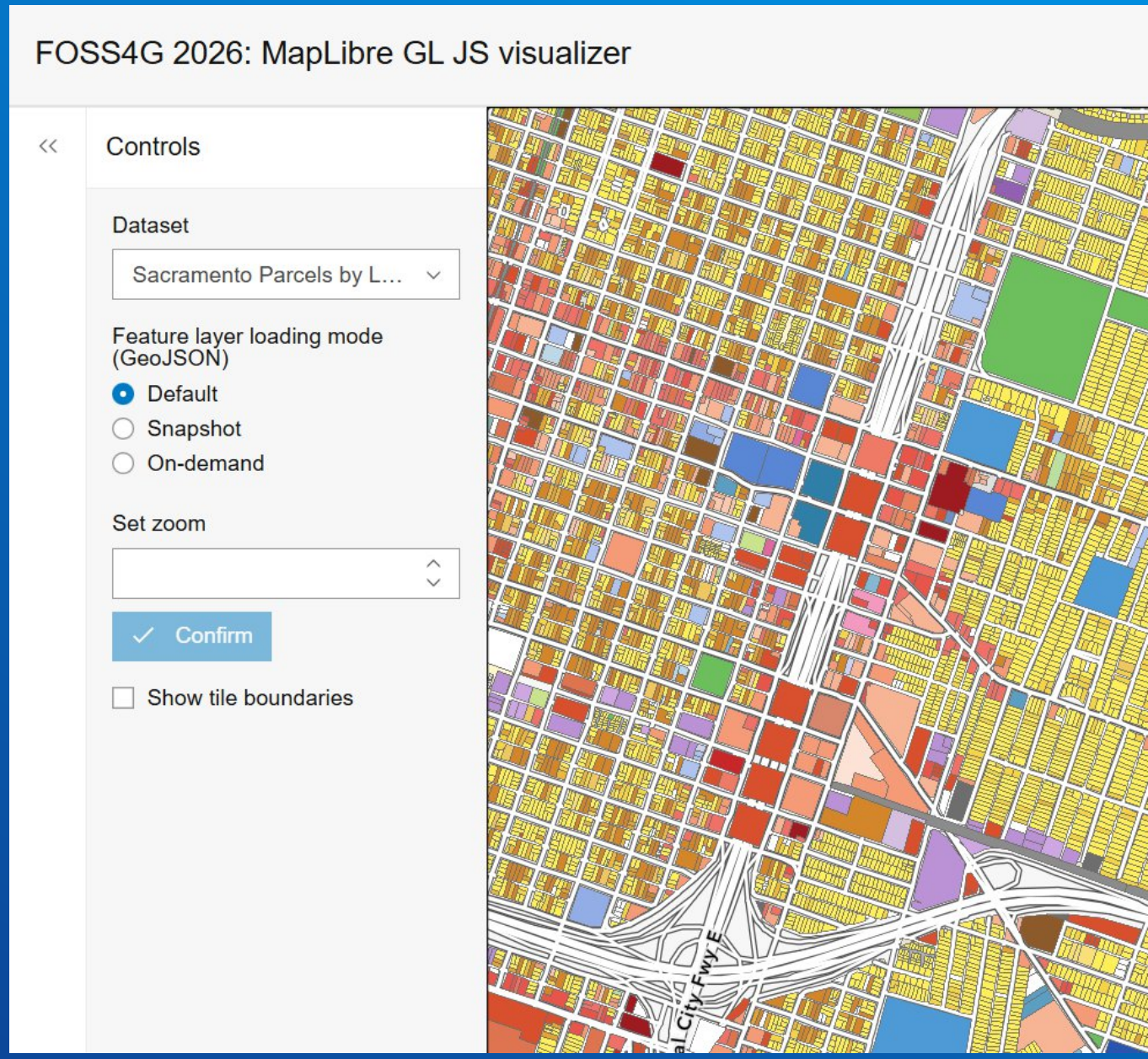
Demo application

Browser application

- Browser-based MapLibre app
- Framework: Svelte, Typescript, Vite

- UI: Calcite Components
- Map: MapLibre GL JS
- Data:
  - Hosting: Vector tile service
  - Streaming: MapLibre ArcGIS plugin

- LINK



Basemap style

ArcGIS Basemap Styles Service

- ArcGIS Human Geography basemap

- All ArcGIS basemaps use MapLibre style JSON

- ~200 style layers of roads, buildings, etc

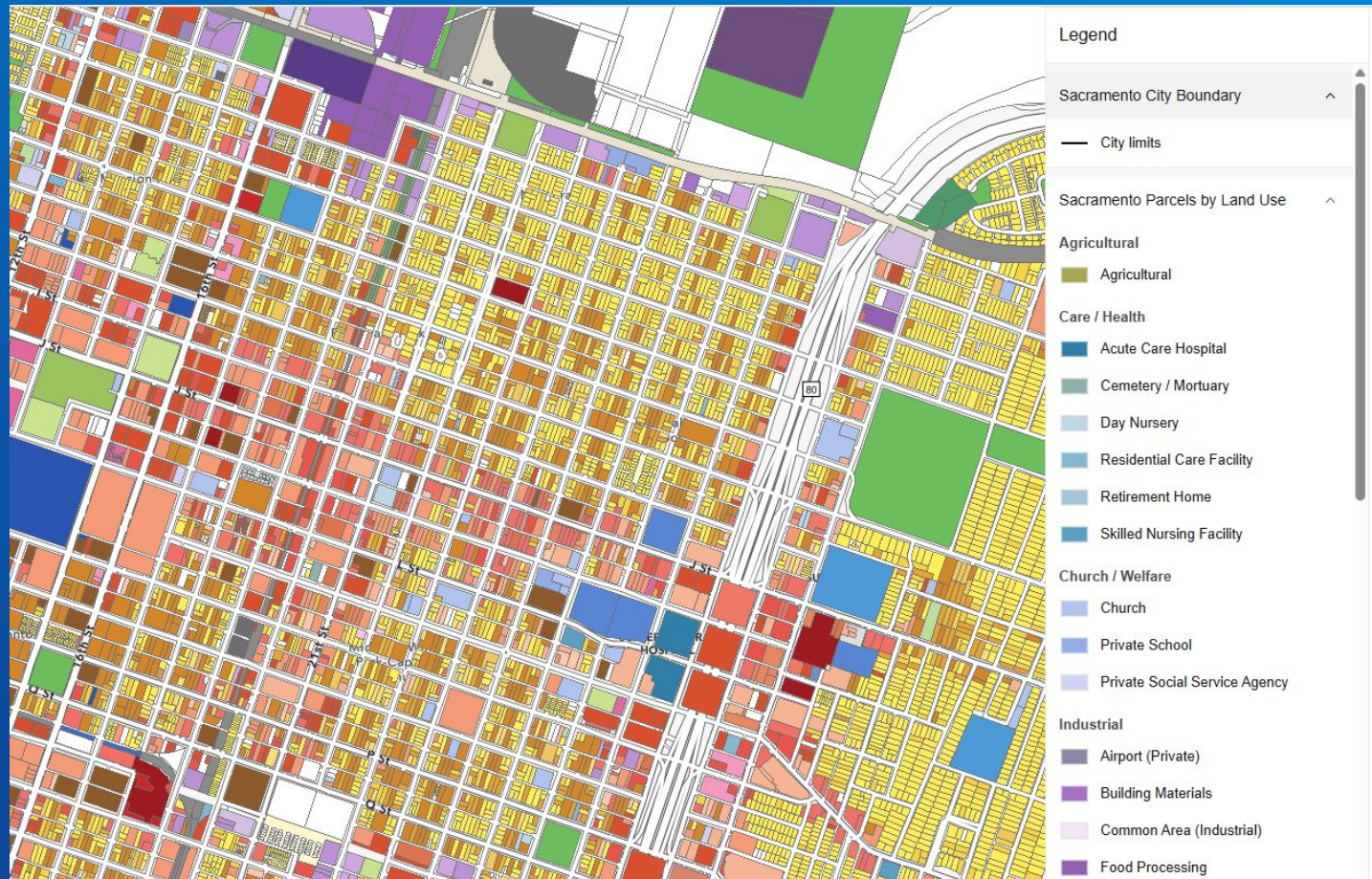
- 50+ other styles available
- Custom basemap support



Dataset 1: Parcels by land use

Chorochromatic style

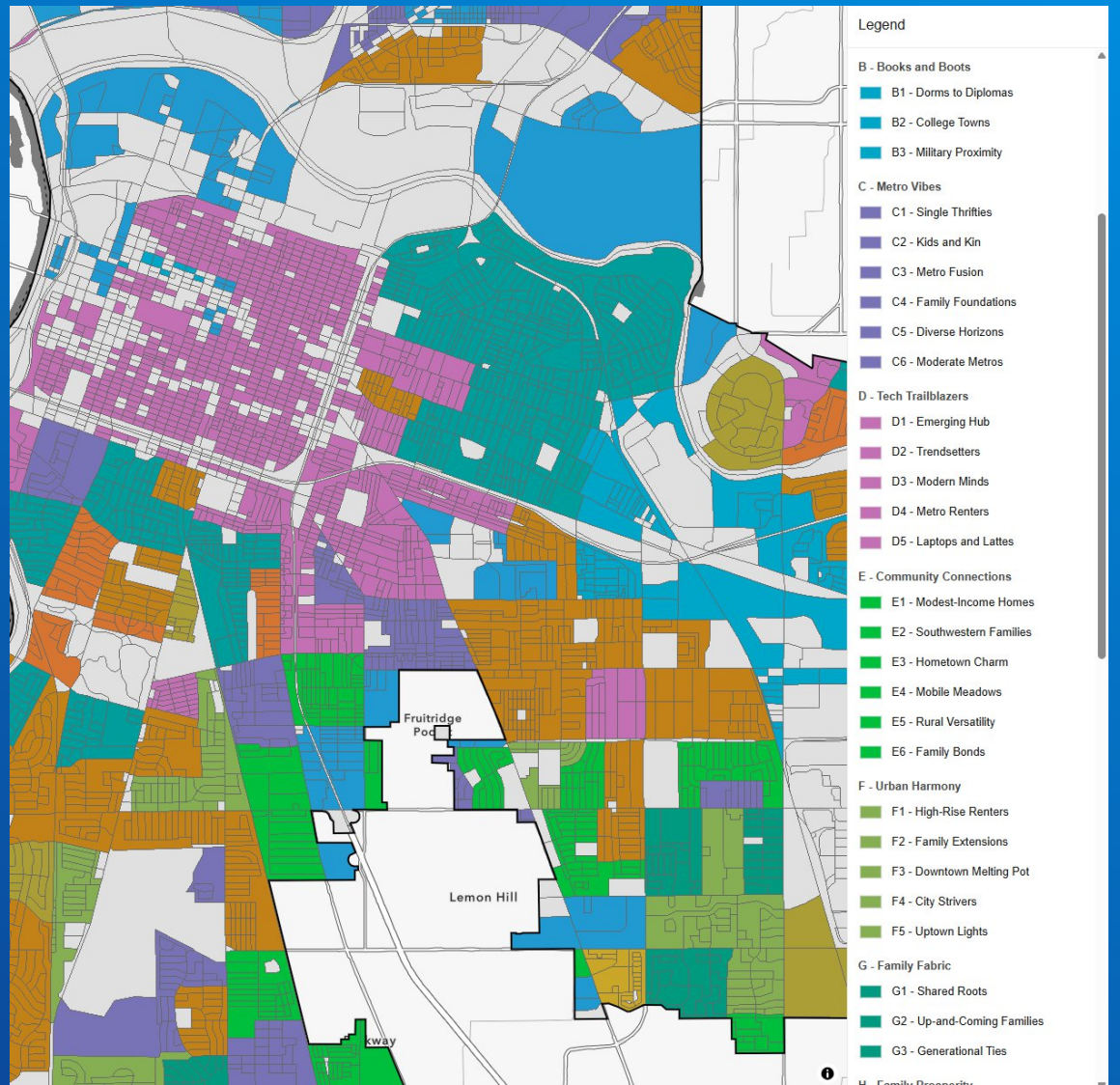
- Access LU_SPECIF property through style spec
- Conditional parcel colors
  - Deep theory behind color choices
- Vector Tiles for optimal performance
- Legend built using Calcite



Dataset 2: Esri Tapestry Segments

Chorochromatic style

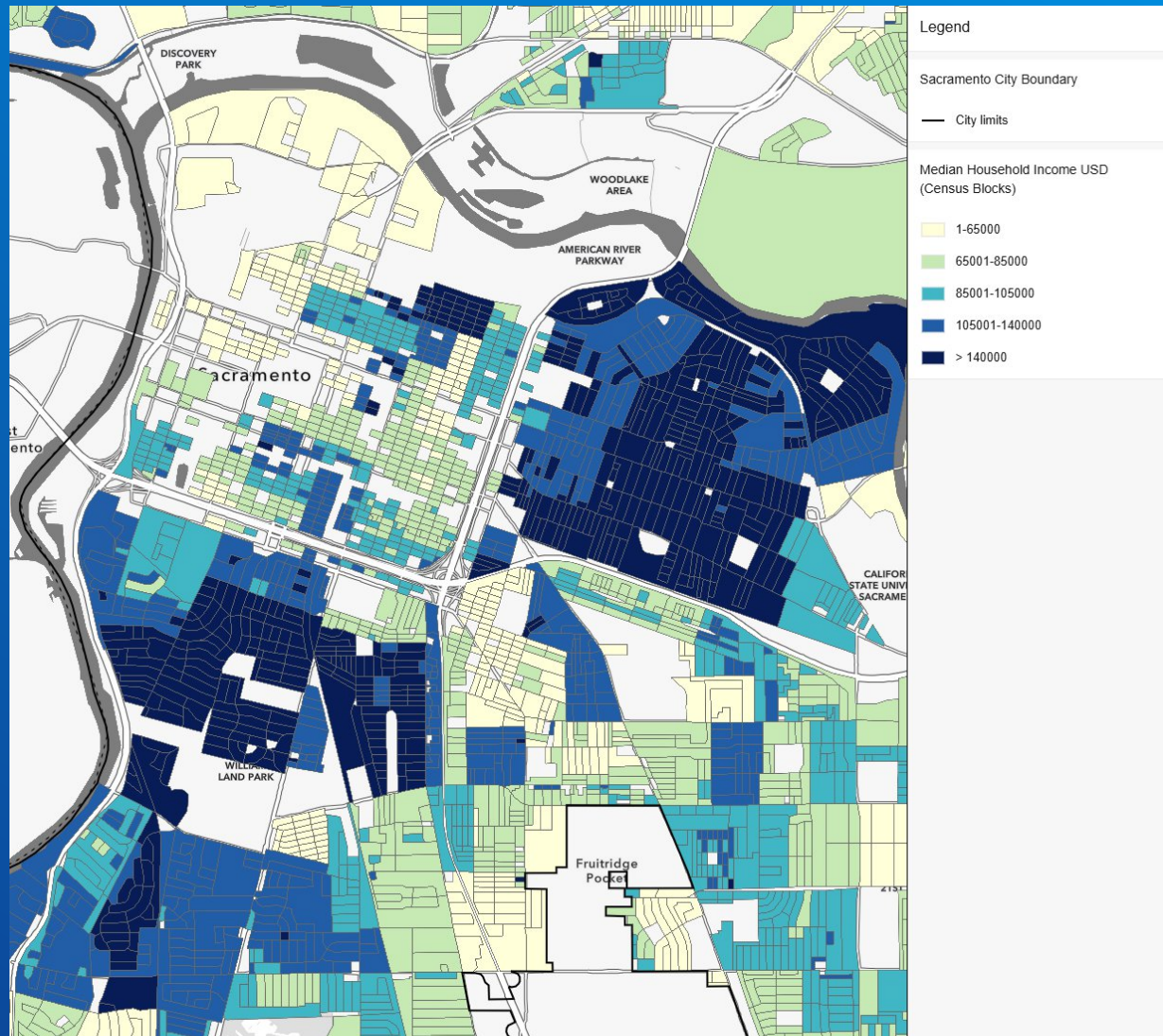
- Neighborhood classification
  - Socioeconomics
  - Demographics
  - Lifestyle characteristics
- “LifeMode” groups are supposed to be useful
- Census block-level data
- Learn more: Tapestry Data



Dataset 3: Median Household Income (USD)

Quintile color ramp

- 5-class Quantile (equal quantity in each class)
- Color has implied meaning
- Census block-level data
- Layer enrichment

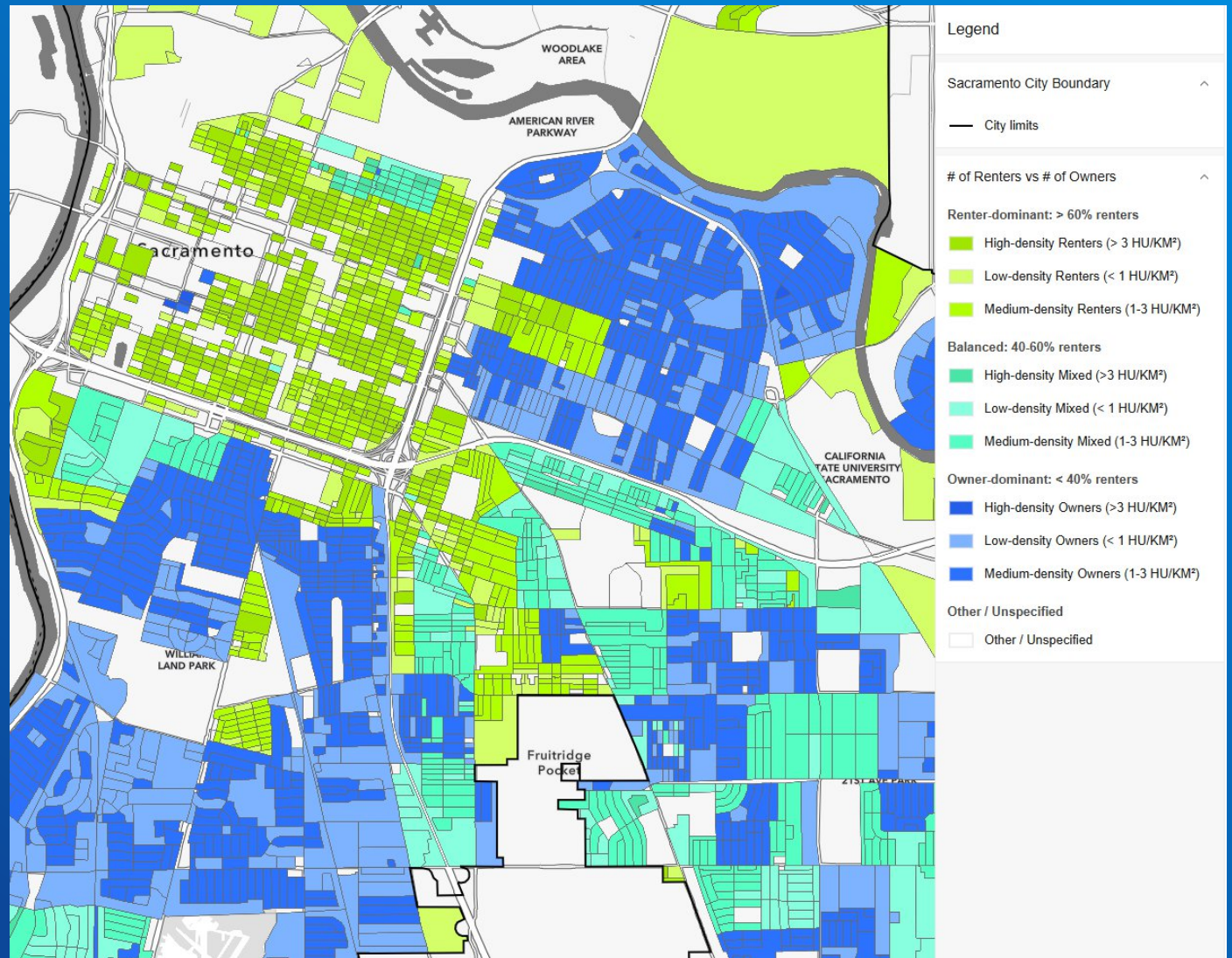


Dataset 4: % of renters vs % of homeowners

Bivariate colors

- Actual variables:
  - Percent of people that are renters (hue)
  - Total household density (saturation)
- Source: Esri Demographics

- Best practice: pre-classify data using an additional attribute

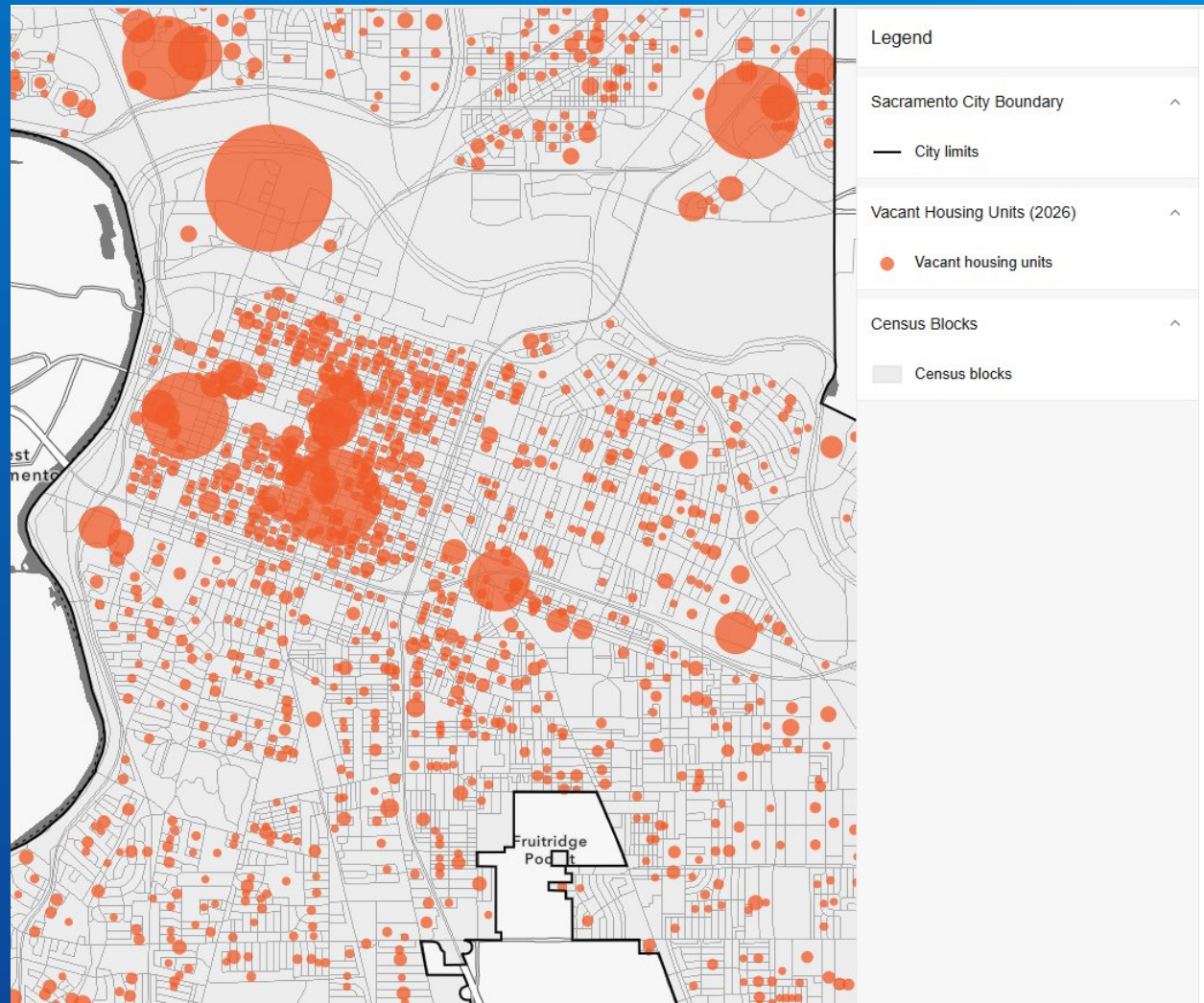


Dataset 5: Number of vacant housing units

Proportional symbols

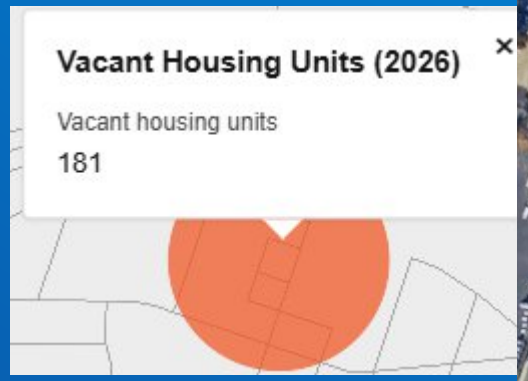
- Points sized by quantity
- Source: Esri Demographics
- Sacramento Housing Element Hub

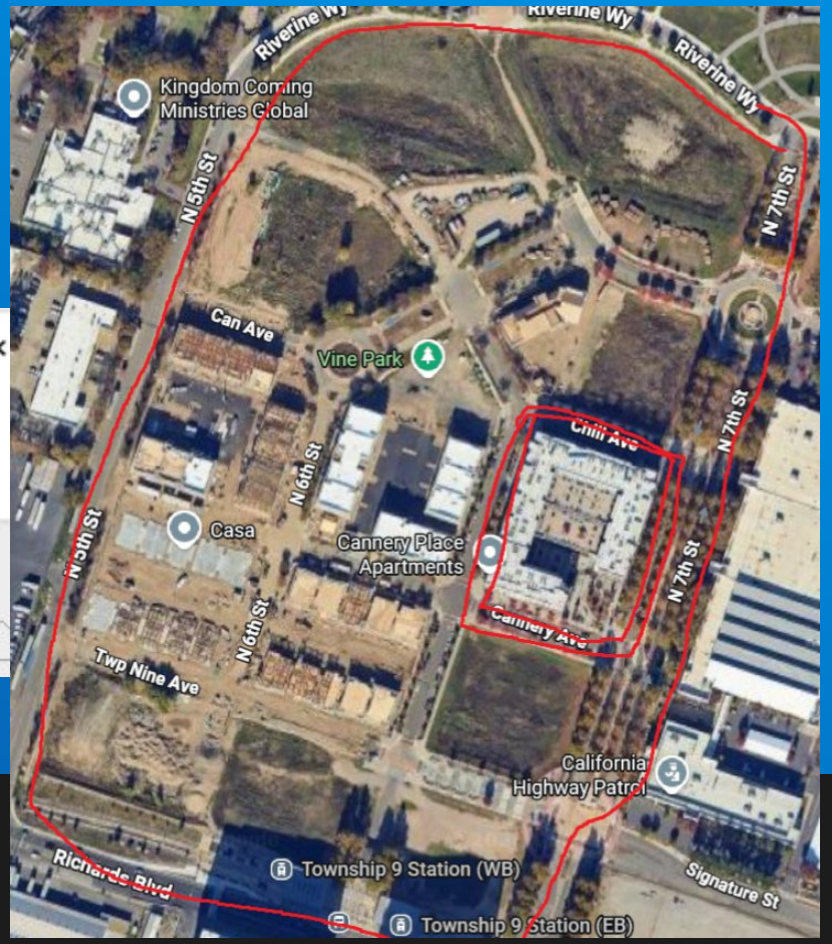
- Highlights specific buildings of interest



The data speaks...

- Yes this is all real data
- https://canneryplaceapts.com/index.html
- https://capitoltowersapartments.com
- “Cheaper to leave it empty”





NEWS

Company Developing Sacramento Riverfront Property Files for Bankruptcy

by: Mark Demsky
Posted: Jun 7, 2017 / 07:25 PM PDT
Updated: Jun 7, 2017 / 07:25 PM PDT

Conclusion & Resources

Takeaways

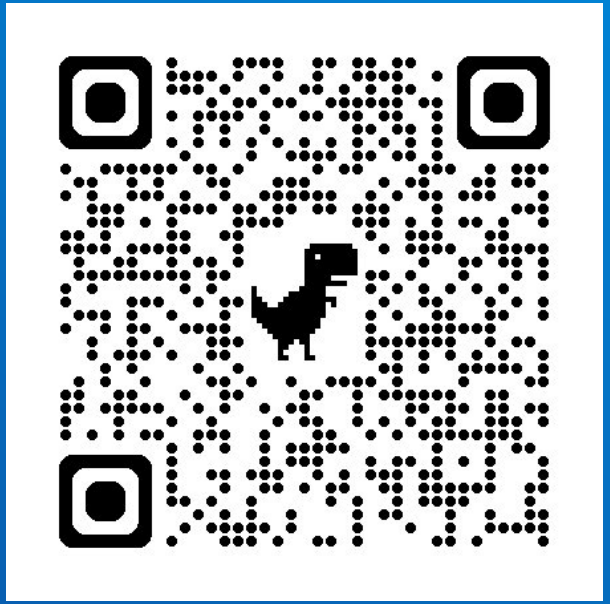
- MapLibre ecosystem!
- Use MapLibre ArcGIS plugin to render open Hub data
- MapLibre style specification + Agentic workflows

Resources

- MapLibre GL JS
- MapLibre ArcGIS plugin
- Data enrichment browser
- Sacramento County Open Data Hub
- Sacramento Housing Element Hub

Styling resources:

- Vector tile style editor
- Maplibre-agent-skills
- gowin20/agent-skills
- APA’s Land Based Classification Standards



Data attribution

- ArcGIS Living Atlas
- Esri Demographics
- City of Sacramento Open Data Hub
- Sacramento County Open Data Hub



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